

Santoshkrishna Tirumala

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SKILLS

Languages: Best in Java and Python; Knowledge of JavaScript+HTML+CSS, TypeScript, C++

Technologies: Linux, Node.js, React, SQL, Git, Docker, Kubernetes, PyTorch, Claude Code CLI, Amazon Kiro CLI, AWS cloud technologies

WORK EXPERIENCE

Amazon Lab126, Sunnyvale

July 2022– Present

Software Development Engineer 2

- Worked on Speech/Audio AI Benchmarking team, which evaluates speech/audio output on various models and devices. **Main areas of focus were Echo devices and Nova Sonic models**
- Onboarded several metrics, including 2 Airflow DAGs w/ AWS EMR, automating previously manual processes, for testing speech-to-speech foundation model capabilities. This exposed areas for improvement in the process, including fundamental model configuration changes. **Measured metrics included speech recognition (ASR) quality, speech synthesis (TTS) quality, and conversational reasoning quality; enabled 20%+ improvement in a model metric for an external leaderboard**
- Implemented scalable API benchmarking/invoke system for 5+ APIs, with features like multithreading and request scheduling. Technologies used include AWS ECS, AWS Batch, AWS SQS, AWS Step Functions, AWS DynamoDB, Python, TypeScript. **This framework enabled metrics on 2 model cards**
- Implemented mobile automation framework, enabling full control of smartphones. Features included opening/closing apps, gracefully handling popups or crashes, and pressing buttons, in a fully extensible manner. Technologies used include AWS Step Functions, AWS DynamoDB, Appium, UIAutomator, Python, Java. **Saved \$200k on manual efforts**
- Implemented smart TV benchmarking framework; allowed manipulation of smart TV remote, capture of on-screen data, and machine analysis of video. Technologies used include home robotics, Raspberry Pi, Arduino, capture cards, AWS Rekognition, AWS Batch, AWS Step Functions, AWS DynamoDB, Python, Java
- Maintained, refactored and improved JavaScript+React+Node centralized portal for the team's services. Also worked as away team on AI org's centralized UI (TypeScript+React+Node), replacing deprecated leaderboard solution
- Led operational improvements such as unit/integration testing coverage, Claude Code skills for automating routine tasks, and improved testing environments in several team/away team services
- Guided 3 interns and 2 full time hires, onboarding them to the team's services

EDUCATION

University of Florida, Gainesville

August 2018- May 2022

B.S., Computer Science